

Question to investigate:

What hidden patterns exist in customer purchasing behavior?

Phase 1: The Loyalty & "Value" Discovery

Our objective is to Identify who most valuable customers are and if "discounts" are actually building loyalty or just attracting "one-timers."

Datasets & Columns: * dim_users: user_id, cltv, orders, savings.

fct_campaigns: user_id, discount_type, redemptions.

Reasoning: Joining cltv with savings allows us to see the Profitability Gap. If a user has high savings but low cltv, they are a "deal hunter." If they have high cltv and low savings, they are "organic loyalists."

Methodology: RFM Analysis (Recency, Frequency, Monetary). We will score users based on how recently they ordered, how often, and their total spend to create distinct tiers (Whales, Loyalists, Hibernating, Lost).

Phase 2: The Digital "Intent" Journey

The objective is to determine the Drop-off points in the app and what triggers a final purchase.

Datasets & Columns: * fct_app_event: user_profile_id, name (event type), created_date, attribute_1 (place_id).

- dim_items: id, title, price.

Reasoning: We need to see the behavior *before* the transaction. Do users view a place 5 times before ordering? Does a specific app_event (like viewing a menu) strongly correlate with an order within 24 hours?

Methodology: Path Analysis / Funnel Analysis. We will map the sequence of events (e.g., app_start -> view_place -> order) to calculate conversion rates and identify friction points.

Phase 3: The "Basket" & Customization Affinity

Discover hidden product pairings and upselling opportunities.

Datasets & Columns: * dim_menu_item_add_ons: category_id, price, select_as_default.

- dim_items: add_on_category_ids, purchases.

- fct_invoice_items: product_id, amount, user_id.

Reasoning: This reveals how people "build" their orders. We want to find if people who buy "Product A" (e.g., a Burger) are 80% more likely to pay for "Add-on B" (e.g., Bearnaisesauce) even if it's not the default.

Methodology: Association Rule Mining (Market Basket Analysis). We will use the "Support and Confidence" metric to find which items or add-ons are frequently bought together.

Phase 4: Campaign Effectiveness & "Cannibalization"

See if marketing campaigns are driving *new* revenue or just discounting what people would have bought anyway.

Datasets & Columns: * fct_campaigns: start_date_time, end_date_time, item_ids, minimum_spend.

- dim_items: purchases, price.

Reasoning: By comparing the purchases count during a campaign period vs. a non-campaign period for the same item_id, we can see the "Lift."

Methodology: Cohort Analysis. We will group users by the first campaign they interacted with and track their spending behavior over the following 3 months to see if that campaign resulted in "sticky" customers.